

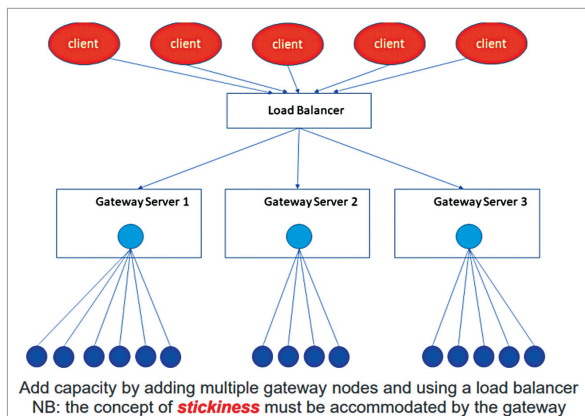


Figure 17: Capacity building by adding gateway nodes

2. Horizontal scaling: This was the approach taken in this project. Multiple instances of gateway services (Figure 17) were run on multiple Raspberry Pi devices and requests distributed across them by a software load balancer. All client requests went to the node running the load balancer.

Using horizontal scaling, in theory, a large number of Raspberry Pi nodes could have been added to the gateway

cluster and very-high concurrent Bluetooth device support achieved. The only issue to be aware of here

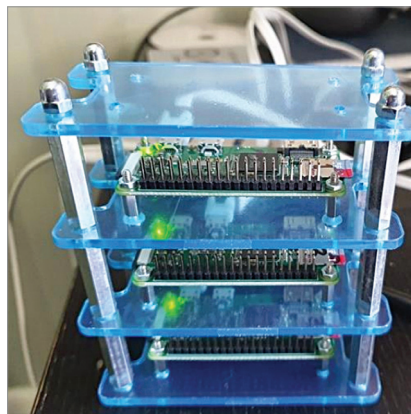


Figure 18: Rack-mounted Raspberry Pi Zero W multi-node solution made by the author

was that there is a load balancer concept known as stickiness. Once a Bluetooth connection has been established from one of the gateway nodes to a particular Bluetooth device, all subsequent requests that relate to that Bluetooth device must be handled by the same gateway node since it holds the Bluetooth connection.

The HAProxy load balancer has excellent support for stickiness, allowing one to configure the behaviour. **END** 🐧

By: Martin Woolley

This article is based on the talk *Understanding reliability in Bluetooth technology* by Martin Woolley, senior developer relations manager, EMEA at Bluetooth SIG during February edition of Tech World Congress and India Electronics Week 2021.

The article was originally published in the May 2021 issue of Electronics For You.

THE COMPLETE MAGAZINE ON OPEN SOURCE



- >> To find dealers to buy copy from news-stand, visit: www.ezine.lfymag.com
- >> To subscribe, visit: pay.efyindia.com
- >> To buy an ezine edition, visit: www.magzter.com choose Open Source For You